Roots & Reptiles: Practice Test 1

Time: 50 minutes. Partners rotate through 15 stations, 3 minutes each, and may not go back or skip ahead. You may use your 2 references (field guides or 1-inch binders), pencils, and up to 2 hand lenses. For every ID, write the common name, and say if it is an NC state symbol when asked.

Partner names: _____ and _____ Score: _____

Station 1: Trees, Part A

1. Count the needles in one bundle of A. Name the tree.

2. Name B. Does it drop its leaves in the fall?

3. Name C. Is its leaf simple or compound?

4. Name D. Write one clue you used.

Station 2: Trees, Part B

5. Name A. Is it an NC state symbol?

6. Name B and C.

7. Name D. Why do many people call it a problem tree?

8. Two riddles: (1) papery, peeling pink-tan bark and grows by rivers; (2) needles 2 or 3 in a bundle, only 3 to 5 inches long, small cones. Name each tree.

Station 3: Vines



9. Name this plant. Is it an NC state symbol?

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10. This vine came from Asia and can grow about a foot a day. It covers trees and blocks their sunlight. What happens to the trees, and why?

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11. Another vine on our list has leaves of three, a hairy vine, and white berries. Name it. Why should you never touch it?

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12. Name the evergreen vine on our list with dark green leaves with 3 to 5 pointed lobes that climbs and covers tree trunks.

Station 4: A Plant That Eats Bugs



13. Name this plant. Is it an NC state symbol?

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14. It makes its own food by photosynthesis. So why does it catch insects?

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15. The trap snaps shut when an insect bumps the trigger hairs. The plant is responding to...

◯ A. light ◯ B. gravity ◯ C. touch ◯ D. water

16. It grows wild only in sunny, wet, sandy places near Wilmington. What could happen if tall shrubs grow up and shade it?

Station 5: A Flower Up Close



17. Name this plant. Is it an NC state symbol?

18. Look at the long, thin parts hanging below the petals with orange-brown tips. The tips hold pollen. What are these parts called?

19. Why are the petals so bright orange?

20. Put these in order: flower, seed, adult plant, pollination, germination, seedling.

Station 6: A Fruit Vine



21. Name this plant. Is it an NC state symbol?

22. The grapes in the photo are still green. What part of the flower did each grape grow from?

23. Where does this plant make its food? What 3 things does it use to make it?

24. Birds and foxes eat the grapes and drop the seeds far away. How does this help the plant?

Station 7: Inside a Seed

25. What is the thin outer skin called, and what is its job?

26. The 2 big halves are the cotyledons. What is their job?

27. Find the tiny baby plant inside. What is it called?

28. When the bean grows, name one job of its roots and one job of its stem.

Station 8: Plant Responses

Read each story. Name the tropism: gravitropism, phototropism, thigmotropism, or hydrotropism.

29. A plant on a windowsill bends toward the window.

30. Tree roots grow toward a leaky water pipe under the ground.

31. A flowerpot tips over on its side. A week later, the stem has curved to grow straight up again.

32. A pea plant's tendril coils around a string it touched.

Station 9: Garden Experiment

Four bean plants started the same. After 3 weeks: Pot 1 (sunny window, watered twice a week) is green and strong. Pot 2 (dark closet, watered twice a week) is tall, thin, and pale yellow. Pot 3 (sunny window, watered every day, water sits on top) has yellow, droopy leaves and smelly soil. Pot 4 (sunny window, 12 seeds in one small pot) has small, skinny plants.

33. Why is Pot 2 pale and thin?

34. What went wrong in Pot 3?

35. Why are the plants in Pot 4 so small?

36. A gardener adds fertilizer to Pot 1. What does fertilizer give a plant? What can happen if you add too much?

Station 10: Big Reptile



37. Name this animal. Is it a reptile or an amphibian?

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38. Give 2 clues that show it is a reptile or an amphibian.

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39. In dry times, alligators dig 'gator holes' that stay full of water. How does this help other animals?

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40. Alligators often lie in the sun on riverbanks. Why?

Station 11: Snakes



41. Name this snake. Hint: look at the snout.

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42. What is its favorite food?

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43. When scared, it flattens its neck and hisses. If that does not work, what does it do?

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44. A thick, dark snake of eastern swamps shows a white inside mouth when it is scared. Name it, and name the harmless snake people often mix it up with.

Station 12: Lizards



45. Name this lizard.

46. Young ones have bright blue tails. If a bird grabs the tail, it breaks off and wiggles. How does this help the lizard?

47. Name the legless lizard on our list. Give one clue that shows it is a lizard and not a snake.

48. Lizards like this eat insects and spiders. How does that help people?

Station 13: Turtles



49. Name this turtle. Is it an NC state symbol?

50. Which turtle on our list is the NC state reptile? What can it do with its shell?

51. Where do turtles lay their eggs, and what do the babies look like when they hatch?

52. Which turtle on our list has a black shell with yellow dots?

Station 14: Salamanders



53. Name this salamander.

54. It has no lungs. How does it breathe, and why must it stay in damp places?

55. It lays its eggs on land under logs, and the babies hatch looking like small adults. How is this different from most frogs?

56. Which salamander on our list is the NC state salamander?

Station 15: Frogs



57. Name this frog.

58. What does a tadpole eat? What does the adult frog eat?

59. Frogs take in water and air through their thin skin. Why do scientists say lots of healthy frogs are a sign of a healthy pond?

60. Which frog on our list is the NC state frog?

27	The embryo.
28	Roots: hold the plant in the soil and take in water and nutrients. Stem: holds the plant up and carries water and food.
Station 8: Plant Responses	
29	Phototropism.
30	Hydrotropism.
31	Gravitropism.
32	Thigmotropism.
Station 9: Garden Experiment	
33	Too little light. It cannot make enough food, so it grows pale and stretches up looking for light.
34	Too much water. The roots cannot get air and start to rot, so the leaves turn yellow and droop.
35	Crowding. The plants compete for water, nutrients, light, and space.
36	Nutrients (like nitrogen) that help it grow. Too much can burn the roots and harm or kill the plant.
Station 10: Big Reptile	
37	American alligator; reptile.
38	Reptile clues: dry, scaly skin (armor plates); breathes with lungs; lays eggs with shells on land (any 2).
39	Fish, turtles, birds, and other animals use the water and food there to survive the dry time.
40	Their body heat comes from their surroundings (cold-blooded), so they bask to warm up.
Station 11: Snakes	
41	Eastern hognose snake.
42	Toads.
43	It rolls over and plays dead.
44	Cottonmouth (venomous); banded water snake (harmless).
Station 12: Lizards	
45	Southeastern five-lined skink.
46	The bird goes after the wiggling tail while the lizard escapes. The tail grows back.
47	Eastern glass lizard. It has eyelids (it can blink) and ear openings; snakes have neither.
48	They eat insect pests in yards and gardens.
Station 13: Turtles	
49	Yellow-bellied slider; no.
50	Eastern box turtle. It can close its shell tight like a box.
51	On land, dug into soil or sand. The babies look like small adults (no tadpole stage).
52	Spotted turtle.
Station 14: Salamanders	
53	Red-backed salamander.
54	Through its moist skin. If its skin dries out, it cannot breathe.
55	Most frogs lay eggs in water, and the young are tadpoles that breathe with gills and then change into frogs.
56	Marbled salamander.
Station 15: Frogs	
57	Southern leopard frog.

58	Tadpole: algae and bits of plants. Adult: insects and other small animals.
59	Pollution passes easily through their skin, so if frogs are healthy and many, the water is probably clean.
60	Pine Barrens treefrog.